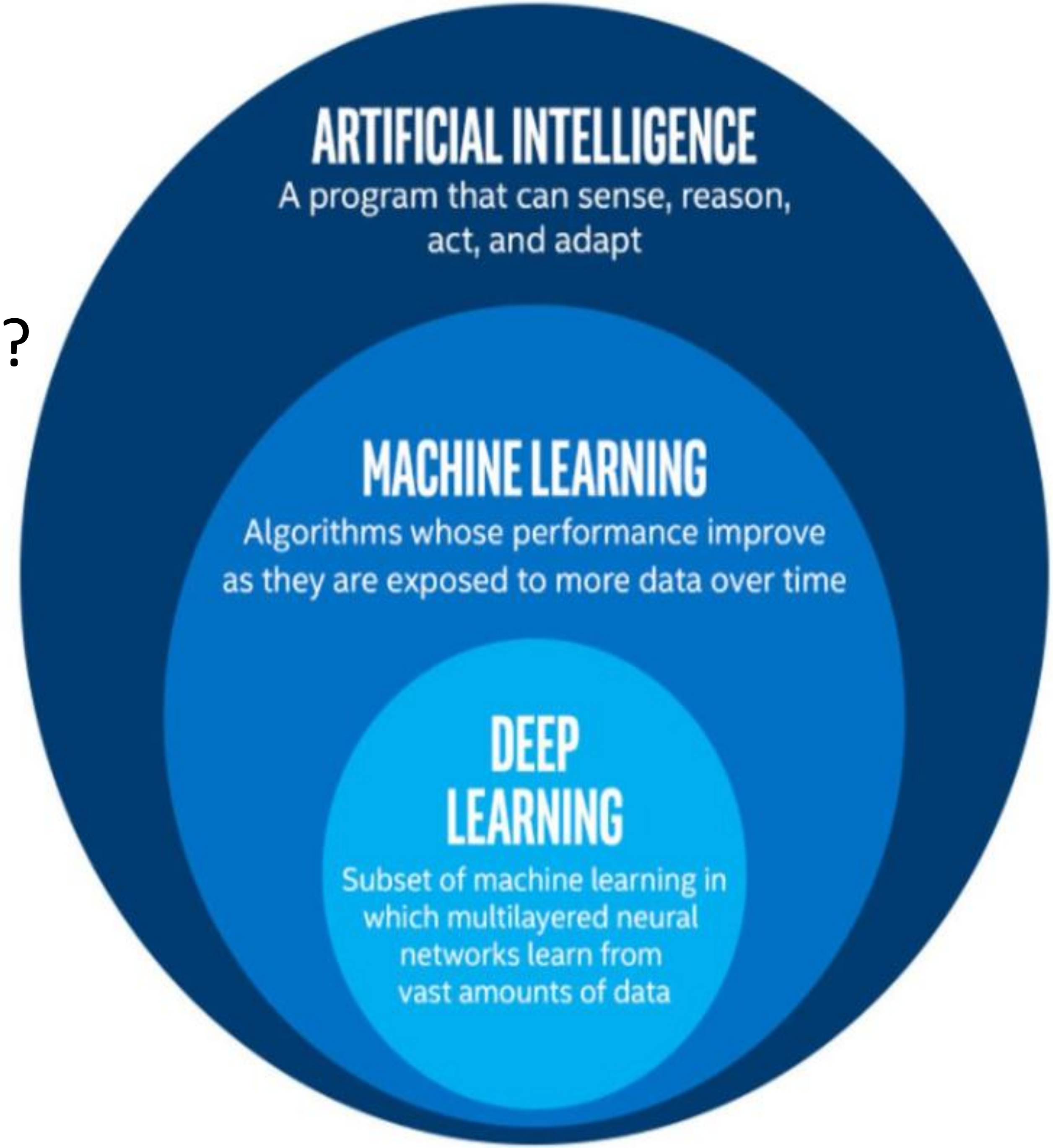
Machine Learning

A comprehensive guide to machine learning with scikit-learn

AI vs ML vs DL?


ARTIFICIAL INTELLIGENCE
A program that can sense, reason, act, and adapt
MACHINE LEARNING
Algorithms whose performance improve as they are exposed to more data over time
DEEP LEARNING
Subset of machine learning in which multilayered neural networks learn from vast amounts of data

What is Machine Learning?

- Machine learning is a method of data analysis that automates analytical model building.
- Using algorithms that iteratively learn from data, machine learning allows computers to find hidden insights without being explicitly programmed where to look.

What it is used for?

- Fraud detection.
- Web search results.
- Real-time ads on web pages
- Credit scoring.
- Prediction of equipment failures.
- New pricing models.
- Network intrusion detection.
- Recommendation Engines
- Customer Segmentation
- Text Sentiment Analysis
- Customer Churn
- Pattern and image recognition.
- Email spam filtering.

What are Neural Networks?

- Neural Networks are a way of modeling biological neuron systems mathematically.
- These networks can then be used to solve tasks that many other types of algorithms can not (e.g. image classification)
- Deep Learning simply refers to neural networks with more than one hidden layer.

Machine Learning

There are different types of machine learning

- Supervised Learning
- Unsupervised Learning

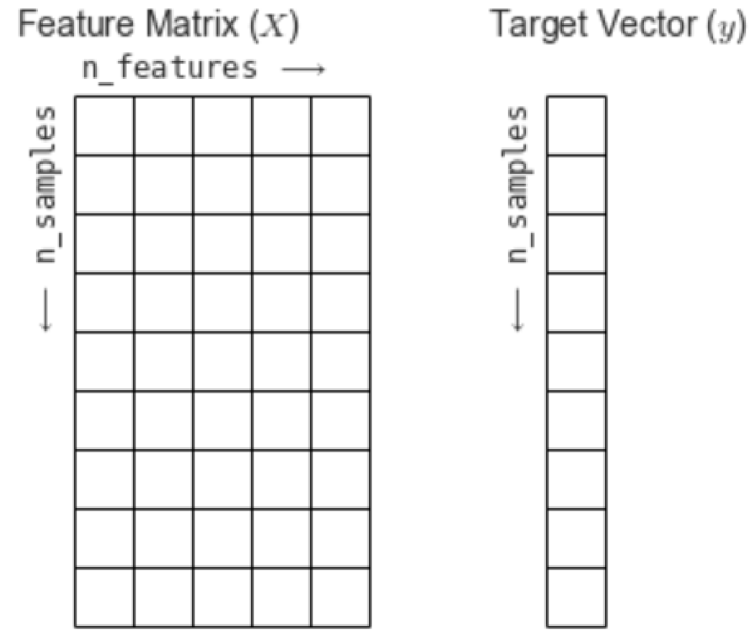
Supervised Learning

- **Supervised learning** algorithms are trained using **labeled** examples, such as an input where the desired output is known.
- For example, a segment of text could have a category label, such as:
  - **Spam** vs. **Legitimate** Email
  - **Positive** vs. **Negative** Movie Review

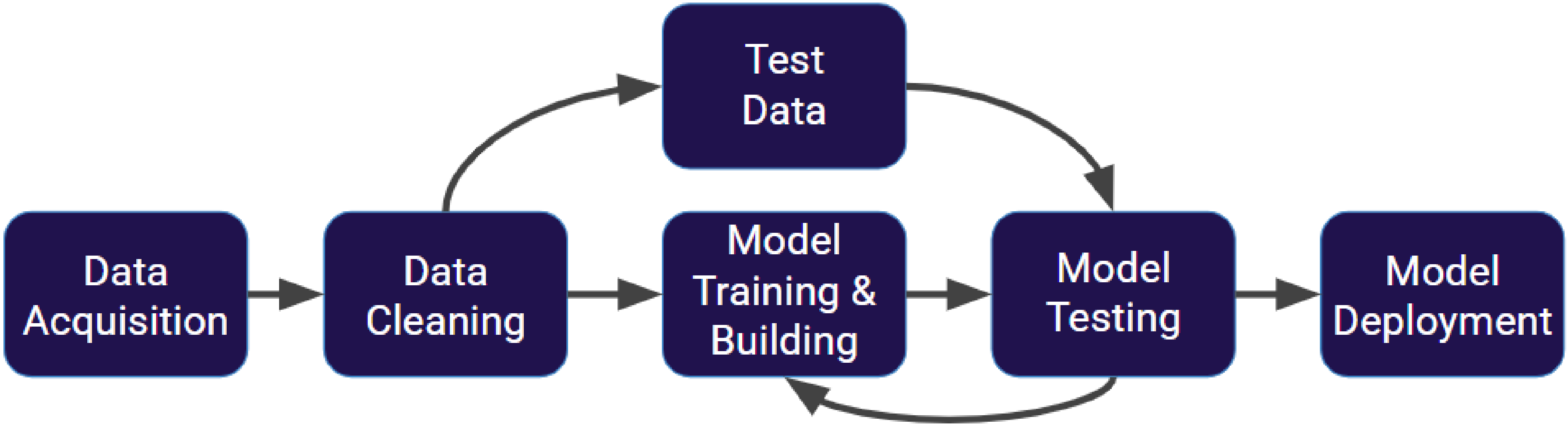
Supervised Learning

Feature Matrix: Set of input columns

Label: Output Column



Machine Learning Process



Supervised Learning

- There are two types of supervised learning
  a) Classification **b) Regression**
- **Classification** is when label column is **discrete** (category) i.e. Fraud detection, Gender detection, Sentiment Analysis
- **Regression** is when label column is continuous (numeric). i.e. Price prediction, Temperature prediction

Supervised Learning

- There are two types of supervised learning
  a) Classification **b) Regression**
    - Linear Models
    - Nearest Neighbors
    - Decision Trees
    - Support Vector Machines
    - Naïve Bayes

Model Evaluation Classification

- The key classification metrics we need to understand are:
  - Accuracy
  - Recall
  - Precision
  - F1-Score

Model Evaluation Classification

$$Accuracy = \frac{TN + TP}{TN + FP + TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Model Evaluation Classification

True	Predicted	Outcome
1	1	**TP**
0	0	**TN**
1	1	**TP**
1	0	**FN**
0	0	**TN**
0	1	**FP**
1	1	**TP**
0	0	**TN**
1	0	**FN**
0	0	**TN**

TP = 3
TN = 4
FP = 1
FN = 2

Model Evaluation Classification

True	Predicted	Outcome
1	1	**TP**
0	0	**TN**
1	1	**TP**
1	0	**FN**
0	0	**TN**
0	1	**FP**
1	1	**TP**
0	0	**TN**
1	0	**FN**
0	0	**TN**

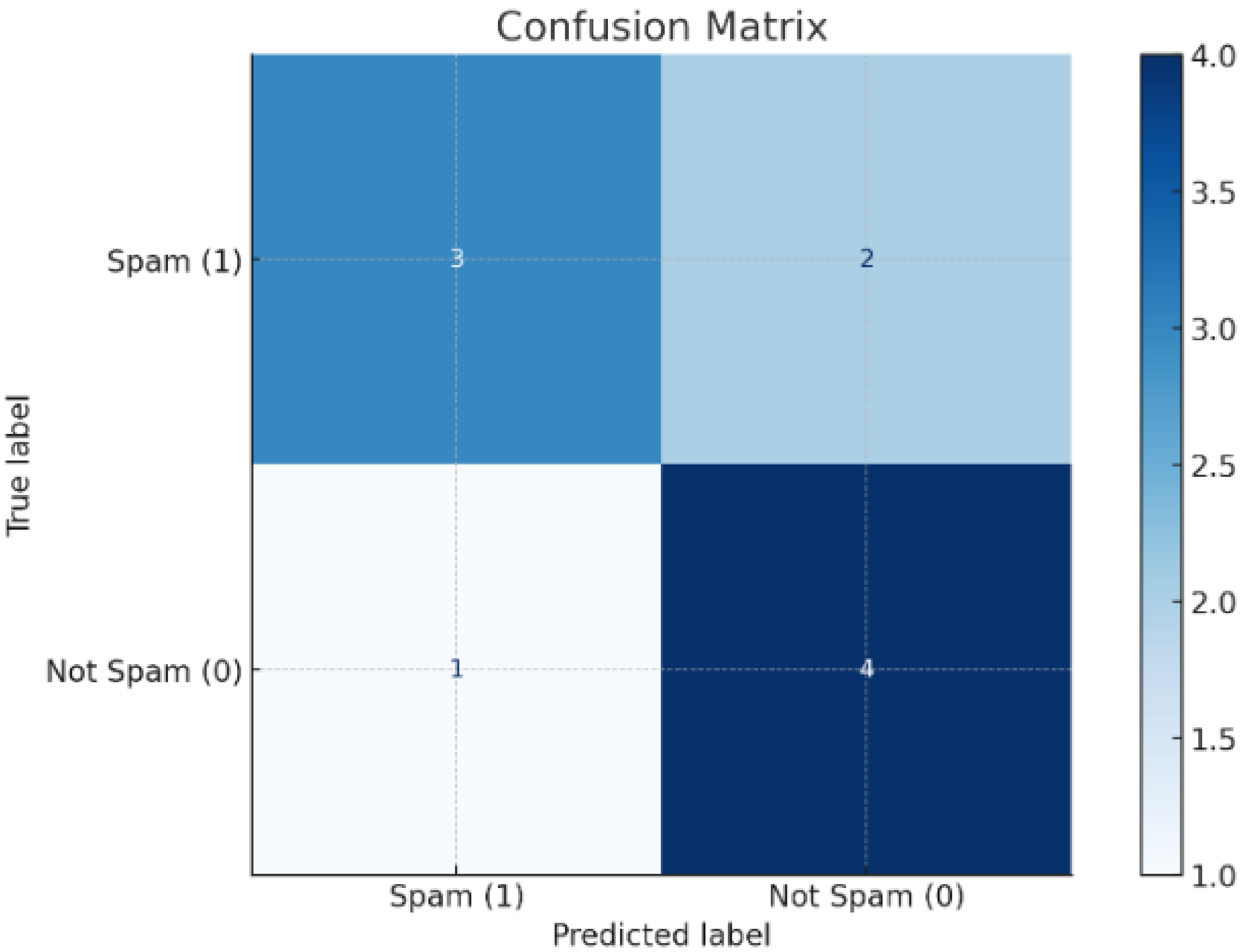
Accuracy = 0.70 (70%)

Precision = 0.75 (75%)

Recall = 0.60 (60%)

F1 Score = 0.67 (67%)

Model Evaluation Classification



Model Evaluation Regression

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$